SDG BLOCKCHAIN ACCELERATOR

Cohort 1 Report

1 Summary

The SDG Blockchain Accelerator is a global initiative that leverages the potential of blockchain technology to accelerate progress toward the Sustainable Development Goals (SDGs). The programme is led by UNDP AltFinLab in partnership with the Blockchain for Good Alliance (BGA), EMURGO Labs, and other technical partners. Cohort 2 is jointly funded by the Cardano Treasury through the Project Catalyst programme and by the Blockchain for Good Alliance.

By applying blockchain solutions to real-world development challenges, the Accelerator promotes transparency, secure digital identities, traceability, and innovative digital public infrastructure. The programme builds on the foundations of the UNDP Blockchain Academy and its Community of Practice of more than 500 members, while integrating the incubation experience of BGA and the technical expertise of EMURGO Labs.

Through this collaborative model, UNDP Country Offices (also referred to as Challenge Owners, COs) and Solution Makers (SMs) are supported with tailored mentorship, technical guidance, and strategic support to design, pilot, and scale blockchain-enabled solutions. The Accelerator ensures that promising initiatives are equipped with the tools, partnerships, and visibility needed to achieve measurable development impact and potential replication across the UNDP ecosystem.

Following the successful implementation of Cohort 1 with the EMURGO track, during Cohort 2 UNDP AltFinLab coordinated programme delivery, communications, and quality assurance, while EMURGO Labs provided technical mentorship, blockchain expertise, and access to its developer community, with a specific focus on solutions built on Cardano. In parallel, UNDP AltFinLab supported teams on internal UNDP procedures, partnership development, resource mapping, and procurement pathways.

2 Cohort 2 Design and Execution

Cohort 2 followed the conclusion of the SDG Blockchain Accelerator Hackathon, which took place over two weeks, from 25 August to 8 September 2025. The hackathon concluded with a Demo Day, during which 49 matched teams composed of Country Offices and Solution Makers presented their pitch decks and solution roadmaps. The pitch deck and the roadmap were the main deliverables of the hackathon phase. These outputs were evaluated by programme partners, including EMURGO Labs and the UNDP AltFinLab focal point. Evaluation was based on predefined criteria covering quality, feasibility, scalability, stakeholder collaboration, and team commitment. Each criterion was scored on a scale from 1 to 5.

Based on the evaluation results, 10 selected teams were invited to join Cohort 2 of the SDG Blockchain Accelerator as part of the EMURGO Labs track. These teams represented diverse regions and institutional contexts:

- 4 teams from the Regional Bureau for Africa (RBA)
- 2 teams from the Regional Bureau for Europe and the CIS (RBEC)
- 2 global teams
- 1 team from the Regional Bureau for Asia and the Pacific (RBAP)
- 1 team from the Regional Bureau for Latin America and the Caribbean (RBLAC)

The selected CO-SM teams were:

- UNDP Ethiopia – TradeChainX
- UNDP Bosnia and Herzegovina – SALA
- UNDP Office of Procurement – ClimaFi
- UNDP Angola – Green Giraffe Zambia
- UNDP Bosnia and Herzegovina – Fuel Switch
- UNDP India – Plastiks
- UNDP Malawi – Xcapit
- UNDP Better Than Cash Alliance – ClimaFi
- UNDP El Salvador – Plastiks
- UNDP Nigeria – Cubid Protocol

Notably, two Solution Makers, ClimaFi and Plastiks, were selected to participate in the Accelerator with two parallel pilot implementations each. Both teams collaborated with two different Country Offices, addressing distinct challenges across different regions and delivering two separate pilot solutions. Prior to confirming their participation under this expanded scope, an additional assessment was conducted to evaluate their operational capacity, resource availability, and commitment to effectively engage in multiple concurrent initiatives. This ensured that both teams were adequately prepared to deliver high-quality outcomes across both pilot tracks.

3 Implementation and Timeline

Following Demo Day on 8 September 2025, selected teams were formally onboarded into Cohort 2 of the SDG Blockchain Accelerator. Participation in the Accelerator provided teams with:

- Four months of tailored mentorship and technical support (September 2025January 2026)
- Global visibility through UNDP and partner networks
- Access to multiple post-acceleration funding opportunities

The teams engaged in an intensive acceleration journey focused on refining, validating, and strengthening their blockchain solutions to ensure alignment with SDG priorities and readiness for realworld pilot deployment. Cohort 2 concluded with a graduation event on February 3, 2026, during which all 10 teams from the EMURGO Labs track successfully graduated. Teams presented their final pitches and solution progress to the EMURGO Labs and UNDP AltFinLab teams.

4 Key Roadmap Components and Deliverables

Throughout the Accelerator, each CO-SM team was required to follow a structured set of deliverables designed to ensure clarity, accountability, and pilot readiness. These deliverables provided both a development framework and a progress benchmark, guiding teams from challenge definition to prototype development and pilot planning.

4.1 Core Deliverables

To ensure consistency, technical soundness, and readiness for pilot implementation, each team was required to submit a set of core deliverables. These documents provided evidence of solution maturity, feasibility, and expected development impact.

1. Roadmap

All teams were required to submit a finalized version of the Roadmap Template, jointly developed and approved by both the Solution Makers and their Challenge Owners. This document serves as the foundational blueprint for pilot implementation in the next phase.

#	Section / Deliverable Title	Description
1	UNDP Challenge Statement refined	Describe the real-world development challenge that the solution addresses. The description should be clear, evidence-based, understandable to external readers, and solution agnostic.
2	Local Context	Explain why this challenge matters in a specific country or regional context. Describe the setting, key beneficiaries, affected communities or sectors, existing efforts to

		address the issue, and the roles of relevant stakeholders.
3	Alignment with UNDP Country Office and National Development Priorities	Explain how the challenge and proposed solution align with the strategic objectives of the UNDP Country Office and relevant national development frameworks. Highlight synergies with UNDP programmes, government strategies, or donor-funded initiatives (e.g. digital transformation, climate resilience, youth employment, financial inclusion).
4	Relevance to UNDP Programmes and Resource Mapping	ndicate whether the pilot builds on an existing initiative or is a new standalone project. If it builds on an existing initiative, describe its scope, scale, timeline, and available resources, including co-financing or in-kind contributions.
5	Key Stakeholders & Potential Partners	List all relevant stakeholders and partners involved in or influencing the project. This may include government agencies, UNDP teams, NGOs, community representatives, private sector actors, and technology providers. Describe their roles and level of involvement.
6	Expected Impact	Outline the intended outcomes from the Country Office perspective. Describe what a successful pilot would enable, such as policy change, improved service delivery, community empowerment, system improvements, or pathways for scaling innovation.
7	Target SDGs and Indicators	Identify up to three specific Sustainable Development Goals and corresponding SDG indicators that the challenge and proposed solution will directly contribute to. Focus on indicators that will be actively addressed.

8	Primary User Persona	Describe the main user or beneficiary of the solution, including their role, context, needs, and challenges.
9	User Story	Expresses user need in the format: "As a [user], I want to [goal], so that [value]."
10	Testing Strategy	Describe how and when user testing will be conducted, including methods, target users, and testing phases during the accelerator.
11	Feedback Loop Design	Explain how feedback from users and stakeholders will be collected, analysed, and integrated into solution improvements.
12	Solution Summary	Provide a brief and clear description of the proposed blockchain-based solution, including its purpose and relevance to the challenge.
13	Core Features and Functionalities	Describe the most important features of the solution and how they address user needs and the identified challenge.
14	Tech Stack and Tools	List the main technologies, platforms, and tools used, including front-end, back-end, data sources, and infrastructure.
15	Blockchain Functionality	Describe the blockchain-specific components of the solution, such as token minting, smart contracts, on-chain verification, credentialing, or data integrity mechanisms.
16	Weekly Workplan	Briefly describe the team's focus during each stage of the programme. This may follow the structure of the hackathon or prototype roadmap, outlining key activities and outputs per week or phase.
17	Solution Demo (PoC)	A working or visual prototype demonstrating the core concept of the solution. The demo will

		be presented virtually to the Country Office team and UNDP or partner focal points.
18	Testing & Feedback Documentation	A summary of user testing activities, including methods used, challenges encountered, and key feedback received.
19	Feedback Integration Summary	A description of how user and stakeholder feedback was incorporated into the solution design and functionality.
20	Pilot-Ready Solution	A stable and functional version of the solution that is ready for pilot deployment.
21	Confirmed Pilot Partnership(s)	Documented confirmation of at least one partner committed to implementing the pilot.
22	Stakeholder Alignment Matrix	A clear overview of stakeholder roles, responsibilities, and coordination mechanisms for the pilot phase.
23	Budget Proposal (Pilot)	A detailed budget for pilot implementation, including estimated costs, units, quantities, and funding sources.
24	Impact Measurement Framework	KPIs, SDG contribution, Monitoring & reporting, Risk assessment & mitigation
25	Post-Acceleration Roadmap & Sustainability Plan	Phased timeline, Budget, Partnerships, Fundraising plan
26	Legal & Compliance Considerations	Summary of key legal, regulatory, or compliance considerations.
27	Visibility Assets / Communication Materials	1–2 slides, visuals, or storylines for use by UNDP or partners to promote your solution.
28	Pitch Deck	Concise and compelling presentation for stakeholders, funders, and partners. ((Please upload to your google drive folder)

29	Final Demo Presentation	An engaging walkthrough of the live solution and its implementation roadmap at the end of the program.

5 Preparation Sessions and Learning Activities

All Solution Makers in the EMURGO Labs track were provided with additional sessions as part of a series of technical and strategic learning sessions:

Technical Sessions

1. Blockchain Fundamentals with a Focus on Cardano Smart Contracts
2. Hands-on Development with Aiken: Building and Interacting with Smart Contracts

Strategic (Non-Technical) Sessions

1. Risk Management and Compliance Requirements
2. Legal Considerations for Blockchain in Development Projects

In addition, EMURGO Labs provided a comprehensive set of self-paced learning modules hosted on Notion. These modules covered topics ranging from blockchain fundamentals to advanced case studies, artificial intelligence, and implementation strategies. Teams were able to access and complete the content independently throughout the programme.

6 Mentorship and Ongoing Support

After acceptance into Cohort 2, each team received continuous support through biweekly coordination and mentorship calls with the UNDP AltFinLab and EMURGO Labs focal point. These sessions focused on progress tracking, strategic alignment, problem-solving, and preparation for pilot implementation.

7 Evaluation and Key Outcomes

Cohort 2 of the SDG Blockchain Accelerator demonstrated strong performance across all stages of the programme, from initial outreach and selection to final solution development and graduation. The scale and depth of engagement are reflected in the following key metrics:

A total of 177 Solution Maker applications were received from more than 40 countries, highlighting significant global interest and diversity of technical expertise. In parallel, over 300 Country Office challenges were reviewed, resulting in the confirmation of 10 Country Offices for targeted matchmaking. Through this process, 12 Solution Maker–Challenge Owner teams were formed and participated in the hackathon phase, engaging in structured co-creation of solution roadmaps.

Following a competitive evaluation process based on predefined criteria, 10 teams were selected to advance to Cohort 2 under the EMURGO Labs track. All 10 Solution Maker–Challenge Owner pairs successfully completed the programme, achieving a 100% graduation rate. Notably, 100% of graduating teams delivered fully developed, implementation-ready roadmaps, demonstrating a high level of maturity and readiness for pilot deployment.

The programme facilitated over 1,000 hours of structured collaboration and co-design activities, including onboarding sessions, hackathons, technical workshops, and iterative roadmap development. In addition, more than 25 stakeholder alignment and validation sessions were conducted, ensuring continuous feedback and alignment with Country Office needs and strategic priorities. Overall, more than 30 Solution Makers and Challenge Owner representatives were actively engaged throughout the programme.

Key outcomes from Cohort 2 include:

- 12 matched Challenge Owner–Solution Maker teams participated in the hackathon Demo Days, of which 10 were selected for Cohort 2 for the EMURGO Labs track.
- All 10 EMURGO Labs teams successfully completed all required deliverables for Cohort 2 graduation, achieving a 100% graduation rate for the EMURGO Labs track.

Cohort 2 demonstrated strong engagement, high-quality outputs, and clear progress toward pilot-ready blockchain solutions aligned with SDG priorities.

8 EMURGO Labs Cohort 2 Details

Matches	Challenge Description	Solution Description	Core Blockchain Features
UNDP Ethiopia & TradeChainX	Ethiopia is advancing its national climate action framework and carbon market	Kijani MRV is a blockchain-backed MRV platform designed to support	Blockchain-Based Carbon Credit Registry and Tokenization, Smart

	strategy, but faces significant challenges in ensuring transparent, reliable Monitoring, Reporting, and Verification (MRV) of emission reductions. Existing MRV processes are largely manual and fragmented, creating risks of data gaps, high verification costs, and limited trust among stakeholders. Without a robust and verifiable digital infrastructure, Ethiopia risks missing opportunities to attract climate finance and participate competitively in global carbon markets. Strengthening trust, traceability, and integrity in carbon credit generation is therefore critical to support the country's emerging carbon market and alignment with national policies and SDG commitments.	Ethiopia's emerging carbon market by enabling transparent tracking, verification, and tokenization of emission reductions. The solution will be piloted over one year across two real-world projects: a clean cooking initiative distributing fuel-efficient cookstoves, and a forestry and coffee landscape restoration project. By digitising MRV processes and tokenising verified carbon credits on Cardano, the platform improves data integrity, reduces verification costs, and enables trusted participation in voluntary and Article 6 carbon markets. The solution supports SDG 13 (Climate Action), SDG 7 (Clean Energy), and SDG 15 (Life on Land), while channeling climate finance to local communities through inclusive benefit-sharing mechanisms.	Contracts for Issuance and Retirement, OnChain Hashing of MRV Evidence, DID-Based Authorization and Verification, Transparent Proof-of-ofRetirement Records
UNDP Bosnia and Herzegovina – SALA	Bosnia and Herzegovina's higher education system	The solution pilots a digital diploma issuance and	On-Chain Credential Fingerprinting, Verifiable Credential

	relies on fragmented governance structures and manual diploma verification processes, which slow student mobility and create opportunities for fraud. Employers and institutions lack a fast, trusted way to verify academic credentials, while universities face legal and operational uncertainty when adopting digital solutions. The absence of a secure, standards-aligned digital diploma infrastructure limits trust, efficiency, and cross-border recognition.	verification platform for the University of Sarajevo, aligned with EU standards and W3C Verifiable Credentials. Students receive digital diplomas in a wallet they control, while employers can instantly verify authenticity through a public portal. Only tamper-proof cryptographic fingerprints are anchored on the Cardano testnet, with no personal data stored onchain, supporting legal alignment and interoperability while complementing existing paper diplomas.	Anchoring, Revocation Status Registry, Tamper-Proof Integrity Proofs
UNDP Office of Procurement – ClimaFi	Many African countries struggle to sustainably operate and maintain hydro-meteorological stations, resulting in unreliable data and weakened early warning systems. Funding for maintenance is inconsistent because donors lack transparent mechanisms to verify data quality and link payments directly to	The Results-Based Payments for Meteorological Data (RBP-MetData) solution introduces a blockchain-based verification and accountability layer that links high-quality weather data to results-based financing. Meteorological data is automatically verified against international standards, generating	Data Receipt Token Minting, On-Chain Hashing of Verified Data, Smart ContractBased Shadow Escrow, Immutable Audit Trail

	results. This "install-and-decay" cycle undermines forecasting accuracy, climate resilience, and trust between data providers and funders.	cryptographic "Data Receipt Tokens" for valid datasets. A smart contract-based shadow escrow demonstrates payfor-results financing, reducing audit costs and creating transparent incentives for sustained data quality.	
UNDP Office of Procurement – ClimaFi	Many African countries struggle to sustainably operate and maintain hydro-meteorological stations, resulting in unreliable data and weakened early warning systems. Funding for maintenance is inconsistent because donors lack transparent mechanisms to verify data quality and link payments directly to results. This "install-and-decay" cycle undermines forecasting accuracy, climate resilience, and trust between data providers and funders.	The Results-Based Payments for Meteorological Data (RBP-MetData) solution introduces a blockchain-based verification and accountability layer that links high-quality weather data to results-based financing. Meteorological data is automatically verified against international standards, generating cryptographic "Data Receipt Tokens" for valid datasets. A smart contract-based shadow escrow demonstrates payfor-results financing, reducing audit costs and creating transparent incentives for	Data Receipt Token Minting, On-Chain Hashing of Verified Data, Smart ContractBased Shadow Escrow, Immutable Audit Trail

		sustained data quality.	
UNDP Angola – Green Giraffe Zambia	Despite improvements in agricultural productivity, smallholder farmers in rural Angola remain excluded from formal and higher-value markets. Structural barriers such as poor aggregation, lack of traceability, quality uncertainty, and dependence on informal middlemen prevent farmers, especially women-led cooperatives, from receiving fair prices. The absence of trusted digital market infrastructure limits income growth, resilience, and inclusive value chain development.	The solution provides a blockchain-enabled agricultural market and traceability platform that connects smallholder farmers directly with verified buyers. Farmers, farms, and plots are digitally identified and GPSmapped, while crop activities and QA-verified data are immutably recorded on Cardano. Verified outputs trigger secure payments and enable end-to-end traceability through processing, supporting food safety, ESG reporting, and inclusive market access.	Immutable Crop and Batch Records, QAVerified OnChain Anchoring, Smart ContractBased Payment Triggers, End-to-toEnd Traceability
UNDP Bosnia and Herzegovina – Fuel Switch	Bosnia and Herzegovina exports electricity to EU markets but lacks the infrastructure to distinguish renewable from non-renewable generation within its grid. With the EU Carbon Border Adjustment Mechanism (CBAM) coming into force, the	Fuel Switch is an Energy Attribute Certificate (EAC) management platform that enables renewable electricity producers to issue, manage, and retire certificates in compliance with international standards. Acting as a service layer on top	On-Chain Certificate Fingerprinting, Tokenized Proof of Issuance, Immutable Audit Metadata, SDGLinked Certificate Records

	inability to verify renewable electricity risks blanket taxation and loss of competitiveness for clean energy producers.	of official registries, the platform automates certificate lifecycles and uses blockchain to provide an immutable audit trail. This ensures transparency, CBAM compliance, and trusted access to international markets.	
UNDP India – Plastiks	Plastic waste recovery relies heavily on informal waste pickers who remain unverified, excluded from digital systems, and unable to access fair economic incentives. At the same time, brands and governments lack trusted mechanisms to verify real recycling outcomes, increasing fraud risks and undermining EPR and ESG reporting.	The solution implements a blockchain-based plastic waste traceability and credit distribution system that digitally records the full lifecycle of plastic recovery. Using QR-coded identities linked to custodial wallets, waste pickers can participate without smartphones or internet access. Verified recovery events generate credits that are automatically allocated across the value chain, ensuring transparency, inclusion, and trusted datasets for EPR and impact reporting.	On-Chain Recovery Event Recording, Automated Credit Allocation Logic, Tamper-Proof Chain of Custody, Fraud and DoubleCounting Prevention
UNDP Malawi – Xcapit	Malawi faces persistent challenges in delivering	The Web3 AgriVoucher solution introduces a	Smart ContractControlled Vouchers,

	development aid transparently and efficiently, particularly in rural areas with limited connectivity. Manual processes, intermediaries, and weak oversight increase the risk of leakage, corruption, and exclusion, undermining donor confidence and the effectiveness of social and agricultural programs.	blockchain-based voucher system combined with an SMS-enabled low-tech wallet. Aid funds are distributed as smart contracts–controlled vouchers that beneficiaries can receive, manage, and redeem via basic mobile phones. The system enables end-to-end traceability, automated rules enforcement, and secure merchant payments without requiring internet access.	UTxOBased Aid Traceability, Time-Locked and Role-Based Spending Rules, Double-Spend Protection
UNDP Better Than Cash Alliance – ClimaFi	A forest carbon project in Oaxaca, Mexico is expected to generate significant carbon credit revenues for thousands of Indigenous community members. However, current manual cash distribution mechanisms create risks of delays, leakage, exclusion, and disputes, threatening trust and compliance with assurance and audit requirements.	The solution provides a blockchainbased payout and compliance rail that ensures carbon revenues are transparently and equitably distributed to all community members. Funds are mirrored onchain through tokenization and governed by smart contracts that enforce community-approved distribution rules. The system integrates with customary governance structures while	Tokenized Digital Twin of Funds, Smart ContractEnforced Distribution Rules, Multisignature Custody, Immutable Audit Records

		providing audit-grade transparency and accountability.	
UNDP El Salvador – Plastiks	El Salvador's recycling ecosystem is constrained by low traceability, weak coordination, and the absence of an operational EPR system. Informal recyclers lack recognition and incentives, while institutions and companies lack credible data to verify recycling outcomes and environmental impact.	The solution is a blockchain-based platform that enables the traceability, verification, and monetization of plastic waste recovery across the national ecosystem. Recovery events are digitally recorded and certified, producing trusted data that can support Plastic Credits, EPR readiness, and sustainability reporting. The platform functions as shared digital infrastructure connecting recyclers, aggregators, government, and sponsors.	Immutable Recovery Records, OnChain Certification Status, Smart Contract–Based Verification Logic, Plastic Credit Provenance
UNDP Nigeria – Cubid Protocol	Rural communities across Nigeria remain underserved by the national electricity grid and depend on costly, unreliable, and polluting energy sources such as diesel generators and kerosene. While solar mini-grids are technically viable, their long-term sustainability is	Solar Village is a blockchain-enabled operations and reporting platform for rural solar mini-grids. It records electricity production, consumption, and payments in tamper-evident ledgers using Cardano Hydra at the mini-grid level, with	Per–Mini-Grid Ledger via Cardano Hydra, On-Chain Anchoring for Auditability, Transparent Energy and Payment Records, Carbon Credit Readiness and Tokenization Pathway

	limited by weak operating models, poor tariff collection, fragmented financial records, and limited transparency. Mini-grid operators struggle to demonstrate predictable revenues, investors face high verification costs, and governments lack reliable data for oversight. The core challenge is the absence of trusted, transparent, and inclusive operational and reporting infrastructure to enable mini-grid scale-up and access to climate and carbon finance.	daily proofs anchored to Cardano mainnet. This creates a permanent audit trail for operators, UNDP, and oversight stakeholders, improving transparency and accountability. The platform also establishes a verifiable data foundation for future carbon credit issuance based on proven kWh generation and use, supporting scale-up and long-term financial sustainability.	

9 Case Studies

9.1 UNDP Ethiopia – TradeChainX

Challenge

Ethiopia is rapidly developing its national climate and carbon market framework, yet it faces a critical gap in robust, transparent, and scalable Monitoring, Reporting, and Verification (MRV) systems. Current MRV processes are largely manual, fragmented, and costly, making it difficult to consistently verify emission reductions, prevent double counting, and build trust among regulators, buyers, and investors. Without a credible digital verification layer aligned with national systems, Ethiopia risks missing opportunities to access high-integrity carbon markets and mobilize climate finance for vulnerable rural communities.

Solution Overview

Kijani MRV is a blockchain-backed digital MRV platform designed to complement Ethiopia's national MRV and carbon registry systems. The solution immutably anchors verified project-level emissions data on the Cardano blockchain, creating a trusted audit trail without replacing government authority. The pilot demonstrates the platform across two priority sectors: forest and landscape restoration under the FOLUR programme, and household clean cooking through eQub's clean cooking initiative. By digitizing verification and linking it to future carbon credit issuance, Kijani MRV strengthens transparency, traceability, and trust across Ethiopia's emerging carbon market ecosystem.

Key Blockchain Features

- On-Chain MRV Evidence Anchoring: Cryptographic hashes of MRV dossiers (satellite data, field evidence, verifier reports) are stored on Cardano, ensuring tamper-proof auditability without exposing sensitive data.
- Tokenized Carbon Credit Registry: Verified emission reductions are represented as on-chain carbon credit tokens, each linked to project origin, vintage, and SDG metadata.
- Smart Contract Issuance & Retirement: Cardano smart contracts govern credit issuance, transfer, and permanent retirement, preventing double counting and fraud.
- Decentralized Identity (DID) Authorization: Verifiers, issuers, and regulators interact with the system using credential-based DIDs, ensuring only authorized actors can approve or issue credits.
- Public Proof-of-Retirement: Human-readable, public verification pages allow buyers and auditors to independently confirm credit retirement and legitimacy.

Expected Impact

- Systemic Impact: Improved integrity, transparency, and auditability of Ethiopia's carbon market infrastructure, aligned with national MRV development and Article 6 readiness.
- Community Impact: Verified climate outcomes across 5,000–10,000 hectares of restored landscapes and 3,000–5,000 clean-cooking households, supporting income generation, health benefits, and reduced deforestation pressure.
- Market Impact: Creation of market-ready, high-integrity carbon assets that attract international buyers and climate finance, strengthening Ethiopia's position in voluntary and future compliance markets.

- SDG Alignment: Direct contributions to SDG 13 (Climate Action), SDG 15 (Life on Land), and SDG 7 (Affordable and Clean Energy), with strong gender and rural livelihood co-benefits.

Prototype & MVP Plan

- Build a pilot-ready MRV platform integrated with mobile data collection, satellite analytics, and Cardano smart contracts.
- Demonstrate end-to-end workflow: project registration → MRV data submission → verification → tokenized credit issuance → simulated transfer and retirement.
- Run controlled testnet simulations for both clean cooking and forestry use cases.
- Conduct usability testing and stakeholder validation sessions with cooperatives, households, verifiers, and government observers.

Expected Outputs

- Functional MRV platform deployed on Cardano testnet.
- On-chain hashes of verified MRV datasets with SDG-linked metadata.
- Tokenized carbon credit issuance and retirement demonstrations.
- Role-based dashboards for implementers, verifiers, regulators, and observers.
- Documented stakeholder feedback and integration report.

Pilot & Scalability

The pilot will run over 12 months across selected FOLUR landscapes and clean cooking communities, following phased onboarding, field testing, verification simulation, and buyer-side review.

- Short term: Validate digital MRV workflows and registry logic within existing UNDP programmes.
- Medium term (2–3 years): Integrate with Ethiopia's national carbon registry and expand to additional sectors such as agriculture and renewable energy.
- Long term: Position Kijani MRV as a national digital public good for carbon market integrity, with replication potential across East Africa and other UNDP country offices.

Business & Sustainability Model

- Programme Integration: Embedded within existing UNDP climate and restoration programmes (FOLUR, clean cooking), avoiding parallel systems.
- Climate Finance Enablement: Future revenues from verified carbon credits help sustain platform operations and community benefit-sharing.

- Government Adoption Pathway: Designed for gradual institutionalization within national MRV and registry frameworks.
- Replicability: Modular architecture allows reuse across sectors, regions, and countries, reducing marginal costs for scale.

9.2 UNDP Bosnia and Herzegovina – SALA

Challenge

Bosnia and Herzegovina's higher-education system relies on fragmented governance structures and manual diploma verification processes that are slow, costly, and vulnerable to fraud. Universities issue paper-based diplomas that are difficult to verify across institutions and borders, limiting student mobility and employer trust. At the same time, public authorities and universities lack a legally aligned, interoperable digital credential infrastructure that complies with EU standards while respecting data protection requirements. This creates inefficiencies, increases administrative burden, and undermines confidence in academic qualifications, particularly relevant as Bosnia and Herzegovina advances its EU integration agenda.

Solution Overview

SALA is a blockchain-enabled digital diploma issuance and verification platform piloted with the University of Sarajevo. The solution issues EU-standards–aligned digital diplomas as W3C Verifiable Credentials that students control in their own digital wallets. Employers and institutions can instantly verify credentials through a public portal, while only a tamper-proof cryptographic fingerprint is anchored on the Cardano testnet, ensuring no personal data is stored on-chain. The solution complements existing paper diplomas and provides concrete inputs for legal and regulatory alignment at the canton and national level.

Key Blockchain Features

- On-Chain Credential Fingerprinting: Cryptographic hashes of issued diplomas are anchored on Cardano, ensuring integrity and tamper-evidence without exposing personal data.
- Verifiable Credentials (VCs): Diplomas are issued as W3C-compliant digital credentials, enabling interoperability across systems and borders.
- Revocation & Status Registry: On-chain anchoring supports real-time verification of credential validity (valid/revoked).
- Decentralized Trust Layer: Blockchain acts as a neutral trust anchor for verification across universities, employers, and institutions.

- Standards Alignment: Designed to align with OpenID4VCI/VP and EU digital credential frameworks.

Expected Impact

- Operational Impact: Faster and more reliable diploma verification for employers and institutions, reducing administrative overhead.
- Governance Impact: Strengthened trust and transparency in higher-education credentials, supporting public sector oversight.
- Student Impact: Increased mobility and ownership of academic credentials through wallet-based digital diplomas.
- Policy Impact: Evidence base for Sarajevo Canton and national authorities to inform legal alignment with EU digital education standards.
- SDG Alignment: Contributions to SDG 4 (Quality Education) and SDG 16 (Peace, Justice and Strong Institutions).

Prototype & MVP Plan

- Implement end-to-end diploma issuance workflow: registrar upload → digital credential issuance → student wallet delivery → public verification.
- Anchor diploma fingerprints on Cardano testnet for integrity verification.
- Demonstrate revocation and real-time status checking through the verifier portal.
- Conduct live demonstrations with university administrators, students, and external verifiers.

Expected Outputs

- Functional MVP for digital diploma issuance and verification.
- On-chain hashes representing issued academic credentials.
- Public verification portal showing credential status (valid/revoked).
- Student wallet integration for credential storage and presentation.
- Documented legal, technical, and stakeholder feedback.

Pilot & Scalability

The pilot focuses on the University of Sarajevo, serving as a controlled environment to validate technical feasibility, usability, and legal considerations.

- Short term: Validate the issuance and verification workflow within a single university.
- Medium term (1–3 years): Expand to additional universities and cantons, and integrate with national education registries.

- Long term: Position the solution as a national digital credential infrastructure aligned with EU frameworks, with replication potential across the Western Balkans.

Business & Sustainability Model

- Institutional Adoption: Universities and ministries adopt the platform as part of standard diploma issuance workflows.
- Cost Efficiency: Reduced verification and fraud-related costs offset system maintenance over time.
- Standards-Based Design: Interoperability reduces vendor lock-in and longterm dependency.
- Replication Potential: Modular architecture enables reuse across education, professional certification, and skills credentialing use cases.

9.3 UNDP Office of Procurement – ClimaFi

Challenge

Many African countries face persistent challenges in maintaining hydro-meteorological infrastructure and sustaining the flow of high-quality weather data needed for early warning systems and climate resilience. Meteorological stations are often deployed through donor-funded projects but fall into an "install-and-decay" cycle due to the absence of predictable, results-based financing for operations and maintenance. Manual reporting, fragmented data verification processes, and limited transparency make it difficult for donors and global initiatives, such as the Systematic Observations Financing Facility (SOFF), to link funding directly to data quality and performance. This creates a critical "missing middle" between infrastructure investment and longterm sustainability, particularly in Least Developed Countries and Small Island Developing States.

Solution Overview

The Results-Based Payments for Meteorological Data (RBP-MetData) solution is a blockchain-enabled digital compliance and accounting layer that links verified meteorological data to results-based climate finance. Acting as a "digital trust machine," the platform automatically verifies weather data against World Meteorological Organization (WMO) GBON/WIS2 standards, issues cryptographic "Data Receipt Tokens" for valid datasets, and simulates results-based payments through a shadow escrow mechanism. The pilot, implemented in Tanzania, complements ongoing SOFF investments by providing transparent, tamper-proof proof of data quality and performance, without introducing regulatory or financial risk during the testing phase.

Key Blockchain Features

- Data Receipt Tokenization: Verified meteorological datasets trigger the minting of on-chain receipt tokens, each embedding a cryptographic hash of the underlying data.
- On-Chain Audit Trail: Token metadata provides immutable proof of data origin, quality, and timestamp, enabling transparent audits by donors and oversight bodies.
- Smart Contract "Shadow Escrow": Blockchain contracts calculate and display theoretical results-based payments based on verified data, without executing live financial transfers during the pilot.
- Verification Gate Logic: Only data that meets agreed GBON/WIS2 quality criteria can generate tokens, preventing payment for low-quality or incomplete data.
- Institutional Identity & Authorization: Organizational wallets and decentralized identifiers authenticate National Meteorological and Hydrological Services (NMHS) as authorized data publishers.

Expected Impact

- Operational Impact: Improved reliability and visibility of national meteorological data streams, reducing downtime and compliance gaps.
- Financial Impact: Demonstration of how verified data can become a "bankable asset," enabling predictable results-based financing for station maintenance.
- Governance Impact: Reduced audit and verification costs for donors through transparent, algorithmic proof of performance.
- Systemic Impact: A replicable model for transitioning from input-based grants to pay-for-results climate finance.
- SDG Alignment: Direct contributions to SDG 13 (Climate Action), SDG 9 (Industry, Innovation and Infrastructure), and SDG 17 (Partnerships for the Goals).

Prototype & MVP Plan

- Deploy a data ingestion and verification pipeline using real meteorological station data.
- Implement automated GBON/WIS2 compliance checks and anomaly detection.
- Mint Data Receipt Tokens on the Cardano testnet for verified datasets.
- Activate a smart-contract-based shadow escrow to simulate results-based payments.
- Demonstrate role-based dashboards for NMHS providers and SOFF/UNDP oversight.

Expected Outputs

- Functioning MVP demonstrating end-to-end flow: data ingestion → verification → token issuance → shadow settlement.
- On-chain receipt tokens with cryptographic hashes and metadata.

- Provider and donor dashboards showing data quality, compliance rates, and simulated earnings.
- Documentation of testing, stakeholder feedback, and system refinements.

Pilot & Scalability

The initial pilot focuses on selected meteorological stations in Tanzania, aligned with the SOFF Investment Phase.

- Short term: Validate technical feasibility and institutional usability through a six-month shadow-payment pilot.
- Medium term (6–18 months): Engage regulators and explore controlled live settlements through a regulatory sandbox.
- Long term: Scale the solution across additional LDCs and SIDS, positioning it as a standard digital compliance layer for global results-based climate finance mechanisms.

Business & Sustainability Model

- Transaction-Based Sustainability: In the post-pilot phase, a small protocol or transaction fee on verified data-linked finance flows can support system maintenance.
- SaaS & Premium Services: Advanced analytics, real-time anomaly detection, and premium audit reports offered to institutional donors and insurers.
- Institutional Adoption: Integration with UNDP, SOFF, and national meteorological services embeds the platform into existing climate finance workflows.
- Donor Confidence & Scale: Transparent, verifiable performance data strengthens donor trust and unlocks larger, longer-term funding commitments.

9.4 UNDP Angola – Green Giraffe Zambia

Challenge

Despite productivity gains driven by initiatives such as UNDP's Kurima Project, many smallholder farmers in rural Angola remain unable to access fair, formal, and profitable markets. Structural barriers, including geographic isolation, high post-harvest losses, limited aggregation and storage capacity, and reliance on informal middlemen, prevent farmers from capturing value from increased production. The absence of trusted traceability, quality verification, and transparent market infrastructure reinforces mistrust between farmers and buyers, disproportionately affecting womenled cooperatives and entrenching rural poverty. The core challenge is the lack of interoperable digital market infrastructure that links production, quality assurance, payments, and buyers in a verifiable and trusted manner.

Solution Overview

Green Giraffe provides a blockchain-enabled agricultural market access and traceability platform that connects smallholder farmers directly with verified buyers. Built on Cardano, the solution assigns digital identities to farmers, farms, and plots using GPS-mapped boundaries, while crop activities and QA-verified data are immutably recorded on-chain. Buyers publish verified demand, farmers subscribe to market opportunities, and payments are securely linked to verified outputs. The platform integrates quality assurance, logistics, ERP systems, and life-cycle assessment tools to ensure end-to-end traceability, compliance with global food safety standards, and transparent, auditable transactions.

Key Blockchain Features

- Immutable Farmer and Batch Records: Crop activities, harvests, and batch data are cryptographically hashed and anchored on Cardano to create a tamper-proof chain of custody.
- QA-Verified On-Chain Anchoring: Only quality-assured inspections, compliance checks, and lab results are written on-chain, ensuring data integrity.
- Smart Contract-Based Reconciliation and Payments: Escrow and settlement logic automatically releases payments based on verified quantities and quality grades.
- Traceable Market Transactions: Blockchain records link farmers, aggregators, processors, and buyers, reducing disputes and fraud.
- Optional Tokenization Layer: Future batch-level tokens enable provenance tracking, certification, and ESG-linked incentives.

Expected Impact

- Income & Livelihoods: Increased farmer incomes through reduced postharvest losses, fair pricing, and access to formal and premium markets.
- Gender & Youth Empowerment: Strengthened participation of women-led cooperatives and creation of new roles for youth as QA agents and digital aggregators.
- Market Trust: Improved buyer confidence through verifiable proof of origin, quality, and sustainability.
- Systems Impact: Establishment of scalable digital public infrastructure for inclusive, market-oriented agriculture.
- SDG Alignment: Contributions to SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 9 (Industry, Innovation and Infrastructure), and SDG 12 (Responsible Consumption and Production).

Prototype & MVP Plan

- Configure the platform within selected Kurima-supported Rural Transformation Centers.

- Onboard farmers, QA agents, aggregators, and buyers into the digital system.
- Demonstrate end-to-end workflow: demand listing → farmer subscription → QA verification → batch reconciliation → escrow-based payment.
- Anchor verified production and transaction data on Cardano testnet and generate traceability and ESG reports.

Expected Outputs

- Pilot-ready digital platform deployed across selected value chains.
- On-chain records of verified farmer activities, QA inspections, and market transactions.
- Role-based dashboards for farmers, QA agents, buyers, and administrators.
- QA compliance summaries, LCA reports, and ESG metrics.
- Documented user testing results and stakeholder feedback.

Pilot & Scalability

The pilot will be implemented within the UNDP Kurima Project across selected provinces, onboarding 50–100 farmers and engaging the full value chain.

- Short term: Validate technical performance and user adoption in one priority crop and region.
- Medium term (6–18 months): Scale to additional Rural Transformation Centers and thousands of farmers nationwide.
- Long term: Replicate the model across Southern Africa, supporting regional agricultural digital transformation and cross-border market access.

Business & Sustainability Model

- Integration with UNDP Programmes: Embedded within Kurima to leverage existing investments and partnerships.
- Revenue Streams: SaaS licensing, marketplace commissions, API integrations, and certification/MRV services.
- Affordability & Inclusion: Cost structures designed to remain accessible to smallholders while ensuring platform sustainability.
- Blended Financing & Scale: Combination of donor funding, private-sector participation, and climate/ESG finance to support long-term expansion.

9.5 UNDP Bosnia and Herzegovina × Fuel Switch

Challenge

Bosnia and Herzegovina is a net exporter of electricity to EU markets but lacks the digital infrastructure to reliably distinguish renewable from non-renewable electricity once it enters

the shared national grid. With the EU Carbon Border Adjustment Mechanism (CBAM) coming into force in 2026, the absence of a verifiable system to certify renewable electricity risks exposing all exports to carbon taxation, threatening national revenue, producer competitiveness, and energy sector stability.

Solution Overview

Fuel Switch delivers a blockchain-enabled Energy Attribute Certificate (EAC) management platform that enables the certification, issuance, transfer, and retirement of renewable energy certificates. The solution acts as a trusted service layer above existing registries, creating an immutable audit trail for electricity origin while remaining compatible with national regulatory structures and EU market requirements. This allows renewable producers and energy traders to demonstrate CBAM compliance using verifiable digital proof.

Key Blockchain Features

- On-chain minting of renewable energy certificates as verifiable digital assets
- Immutable audit trail linking generation data, ownership, transfer, and retirement
- Cryptographic hashing of verified off-chain data anchored on Cardano
- SDG-linked metadata embedded in on-chain transactions
- Registry-agnostic design enabling interoperability with EU-aligned systems

Expected Impact

- Protection of renewable electricity exports from CBAM-related taxation
- Increased competitiveness of renewable energy producers in EU markets
- Improved transparency and trust for regulators, buyers, and international partners
- Stronger incentives for renewable energy investment and market growth
- Alignment with SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action)

Prototype & MVP Plan

- Demonstration of end-to-end EAC lifecycle: verification, issuance, transfer, and retirement
- On-chain minting of certificate fingerprints on Cardano testnet
- Integration of SDG-linked metadata into blockchain records
- Functional platform interface for certificate management
- Validation through structured stakeholder feedback sessions

Pilot & Scalability

- Pilot deployment with a selected renewable energy producer and buyer
- Validation of technical, regulatory, and market feasibility during the accelerator

- Expansion across the Federation of Bosnia and Herzegovina and national energy market
- Replication potential across Western Balkan countries facing CBAM compliance challenges

Business & Sustainability Model

- Producer onboarding and service fees
- Per-certificate transaction fees for issuance, transfer, and retirement
- Long-term sustainability driven by commercial adoption and regulatory alignment
- Potential integration with regional carbon and climate finance mechanisms

9.6 UNDP India – Plastiks

Challenge

India's plastic waste recovery system relies heavily on informal waste pickers who play a critical role in collection and recycling but remain invisible, unverified, and excluded from digital, financial, and policy benefits. Current waste management processes lack end-to-end traceability, making it difficult for governments and brands to verify genuine recycling outcomes and prevent fraud. Fragmented, largely offline systems exclude waste pickers from fair value distribution while undermining Extended Producer Responsibility (EPR), CSR, and ESG reporting. The core challenge is the absence of a transparent, inclusive, and verifiable digital infrastructure that recognizes informal workers while producing trusted recycling data at scale.

Solution Overview

Plastiks delivers a blockchain-based plastic waste traceability and credit distribution system that records the full lifecycle of plastic, from collection by informal waste pickers to final recycling. Using QR-coded digital identities linked to custodial wallets, the solution enables participation without requiring smartphones, internet access, or digital literacy. Verified recovery events are immutably recorded on blockchain, credits are generated per kilogram of recycled plastic, and benefits are distributed automatically according to predefined rules. The platform produces auditable, policyready datasets that support EPR implementation, sustainability reporting, and inclusive circular economy models.

Key Blockchain Features

- QR-coded digital identities linked to custodial blockchain wallets
- Immutable recording of verified plastic recovery and recycling events
- Smart contract-based automation of credit generation and allocation
- On-chain certification and proof-of-recovery status

- Cryptographic prevention of double counting and fraud
- Blockchain-anchored audit trails supporting EPR, CSR, and ESG reporting

Expected Impact

- Formal recognition and visibility of informal waste pickers
- Improved income stability through transparent, automated benefit distribution
- Fraud-resistant, auditable recycling data for governments and brands
- Policy-ready evidence to support EPR implementation in India
- Scalable and replicable model for national and regional deployment
- Contributions to SDG 12 (Responsible Consumption and Production), SDG 8 (Decent Work), and SDG 13 (Climate Action)

Prototype & MVP Plan

- Deploy an end-to-end pilot covering waste picker → aggregator → recycler flows
- Issue QR-coded digital IDs and onboard custodial wallets
- Record real recovery transactions and recycler proof-of-recycling uploads
- Generate and allocate plastic credits based on verified kilograms
- Anchor recovery and certification records on blockchain
- Validate workflows through weekly coordination and stakeholder feedback

Expected Outputs

- Pilot-ready blockchain traceability and credit distribution platform
- On-chain records of verified recovery and recycling events
- Dashboards for UNDP, NGOs, aggregators, and recyclers
- Auditable datasets for EPR, CSR, and ESG reporting
- Testing and feedback documentation informing system refinement

Pilot & Scalability

- Short term: Pilot deployment with selected NGOs, aggregators, recyclers, and waste picker groups
- Medium term: Expansion to additional cities, waste streams, and partners, increasing verified volumes
- Long term: Integration into national EPR frameworks and replication across other regions and countries

Business & Sustainability Model

- Volume-based platform usage fee linked to verified kilograms of plastic processed
- Tiered pricing with volume discounts to incentivize scale

- Revenue aligned with verified impact ("pay-for-verification" model)
- Long-term sustainability through EPR adoption, corporate demand, and donor co-financing
- Low-friction model with no per-user subscriptions, supporting inclusive scale

9.7 UNDP Malawi – Xcapit

Challenge

Malawi faces persistent challenges in the transparent and effective distribution of development aid and public funds, particularly in rural areas where digital and financial access is limited. Aid delivery processes rely heavily on manual recordkeeping, cash-based disbursements, and centralized intermediaries, creating significant risks of inefficiency, corruption, duplication, and fund diversion. These weaknesses undermine donor confidence, reduce accountability, and prevent assistance from reliably reaching intended beneficiaries, including smallholder farmers, women, and vulnerable households. With over 70% of the population living below USD 2.15 per day and approximately 72% lacking internet access, conventional digital payment solutions are not viable, highlighting the need for a lowtech, transparent, and inclusive aid delivery infrastructure.

Solution Overview

Xcapit delivers a blockchain-enabled AgriVoucher and social assistance distribution system built around an SMS-based low-tech wallet and Cardano smart contracts. The solution enables NGOs and public agencies to create, distribute, and manage aid vouchers with full traceability while allowing beneficiaries to receive and spend assistance using basic feature phones without internet access. Funds are distributed as time-bound, rule-based vouchers governed by smart contracts, ensuring that only eligible recipients can redeem them within defined conditions. By combining on-chain accountability with offline accessibility, the system bridges the digital divide while restoring trust, transparency, and operational efficiency in aid delivery.

Key Blockchain Features

- Smart contract–controlled vouchers implemented as individual UTxOs on Cardano
- Time-locked and identity-bound spending conditions enforced on-chain
- Cryptographic verification of beneficiary and steward roles
- Immutable on-chain audit trail for voucher creation, redemption, and recovery
- Prevention of double spending through Cardano's EUTxO model
- Secure fund recovery mechanisms for expired or unclaimed vouchers

Expected Impact

- Improved transparency and accountability in aid and subsidy distribution
- Reduced corruption, leakage, and operational inefficiencies
- Financial inclusion for unbanked and offline populations via SMS wallets
- Increased donor confidence through auditable, tamper-proof records
- Strengthened trust between citizens, government, and development partners
- Replicable model for agricultural aid and social protection programs across Africa

Prototype & MVP Plan

- Deploy an MVP integrating SMS-based wallet onboarding and voucher issuance
- Implement Cardano smart contracts governing voucher rules and expiry
- Enable NGO partner portal for bulk beneficiary onboarding and disbursement
- Test merchant redemption flows using OTP-based SMS authentication
- Validate system performance on Cardano testnet and prepare pilot-ready deployment Expected Outputs
- Functional SMS-based low-tech wallet for beneficiaries
- Smart contract–based voucher system deployed on Cardano
- NGO-facing dashboard for monitoring disbursements and redemptions
- On-chain records of aid allocation, usage, and recovery events • Technical documentation and testing reports supporting pilot readiness

Pilot & Scalability

- Short term: Pilot deployment with 200–500 beneficiaries in Malawi (e.g. Dowa District) focusing on agricultural aid and social cash transfers
- Medium term: Expansion to additional districts, increased beneficiary volumes, and improved merchant coverage
- Long term: Replication across multiple UNDP country offices using a standardized deployment toolkit and shared reporting framework

Business & Sustainability Model

- Programme-based B2B model targeting UN agencies, NGOs, and implementing partners
- One-time onboarding and configuration fees per country deployment
- Ongoing platform subscription for administration, monitoring, and reporting
- Transaction-based fees aligned with beneficiary and disbursement volumes
- Long-term sustainability through donor funding, programme budgets, and multi-country scale efficiencies

9.8 UNDP Better Than Cash Alliance – ClimaFi

Challenge

Indigenous communities managing large forest carbon projects in Mexico face significant risks in distributing carbon credit revenues transparently and equitably. In San Miguel Chimalapa, approximately USD 3.25 million per year is expected to flow through a single communal account, but current cash-based processes create delays, leakage, governance disputes, and exclusion, particularly for women. The absence of a trusted, auditable payout mechanism undermines community trust, threatens project certification, and increases ESG and assurance risks for buyers and donors.

Solution Overview

The solution is a blockchain-based compliance and payout rail that ensures carboncredit revenues reach every intended community member in a transparent, auditable, and inclusive manner. Built on Cardano, it mirrors fiat funds with on-chain records and executes community-approved distribution rules through smart contracts, while integrating with existing usos y costumbres governance structures. The system functions as digital public infrastructure that reduces risk, enforces fairness, and strengthens long-term climate finance credibility.

Key Blockchain Features

- Custodied Digital Wallets: Platform-managed wallets created for each landowner, Designed for unbanked and low-digital-literacy users
- PIN-Based Authorization: Beneficiaries approve transactions via simple PIN entry, Prevents unauthorized or coerced withdrawals
- Smart Contract Governance: Community-approved distribution rules encoded on-chain, Eliminates discretionary manipulation at execution stage
- Multisignature Security: Large disbursements require multiple approvals (community + agent)
- On-Chain Audit Trail: Every registration, approval, and payout logged immutably, Publicly verifiable without exposing personal identities
- Privacy Shield: Personal data stored off-chain; on-chain records remain pseudonymous

Expected Impact

- Improved Transparency: Full visibility of carbon revenue distribution for all stakeholders
- Faster Payments: Reduction of payout cycles from 30–60 days to under 5 days
- Gender Inclusion: Technical enforcement of ≥45% payouts directly to women landowners

- Stronger Governance: Reduced disputes and increased legitimacy of community leadership
- Lower ESG and Assurance Risk: Evidence-grade auditability aligned with Climate Action Reserve requirements
- Climate Protection: Strengthened incentives for forest conservation by securing financial trust

Prototype & MVP Plan

- Demonstrate end-to-end payout flow:
  - Beneficiary registration and wallet creation
  - PIN setup and consent-based transaction approval
  - Multisig cash-out simulation
- Produce:
  - On-chain transaction hashes with metadata
  - Basic beneficiary and admin dashboards
  - Public audit trail demonstration
- Validate usability through community co-creation sessions

Expected Outputs

- Functional MVP deployed on Cardano testnet
- Smart contracts implementing distribution and security logic
- Web-based transparency dashboard
- Documented user feedback from community leaders and beneficiaries
- Recorded demo and technical documentation

Pilot & Scalability

- Pilot Phase: Initial deployment with ~50 beneficiaries (50% women), Integration with local cash-out agents, Live testing of multiple payout cycles
- Short Term: Scale to all ~4,500 landowners in San Miguel Chimalapa
- Medium Term: Replication across other Carbon Plus forest projects in Mexico
- Long Term: Adaptation for other climate finance flows, renewable energy payments, and government social programs across Latin America

Business & Sustainability Model

- Cost-Recovery Service Model: Platform fees paid by project operators, not beneficiaries
- Risk Reduction Economics: Lower assurance and audit costs for carbon project developers

- Revenue Streams: Program administration fees per payout cycle, Operator subscriptions for dashboards and hosting, Licensing fees for replication in new communities
- Long-Term Viability: Transition from grant-funded pilot to self-sustaining service aligned with climate finance flows

9.9 UNDP El Salvador – Plastiks

Challenge

El Salvador faces significant challenges in solid waste management, including low recycling rates, lack of traceability, and the absence of an operational Extended Producer Responsibility (EPR) system. Plastic recovery depends largely on informal grassroots recyclers who work in precarious conditions without formal recognition or fair economic incentives. The core challenge is enabling transparent, verifiable, and scalable systems for plastic waste traceability, certification, and monetization while improving coordination among public institutions, private sector actors, and community stakeholders.

Solution Overview

The solution is a blockchain-based digital platform that enables end-to-end traceability, verification, and monetization of plastic waste recovery across El Salvador's recycling ecosystem. Plastic recovery activities are digitally recorded, verified, and anchored on blockchain, creating immutable and auditable data that can be trusted by government, private sector, and development partners. The platform supports the certification of verified recovery and enables Plastic Credit generation, strengthening EPR readiness, corporate sustainability reporting, and access to climate finance, while remaining inclusive and accessible for grassroots recyclers.

Key Blockchain Features

- On-chain recording of verified plastic recovery events to create immutable audit trails
- Blockchain-based certification of recovery status (verified, certified, rejected)
- Smart contract–enabled verification logic to enforce standardized workflows and roles
- Prevention of double counting and fraud through unique identifiers and onchain anchoring
- Plastic Credit enablement through linkage of certified recovery data to credit issuance
- Time-stamped, auditable records supporting EPR compliance, ESG reporting, and donor assurance

Expected Impact

- Generation of transparent, auditable data on plastic waste recovery
- Formal recognition and economic empowerment of grassroots recyclers
- Improved coordination between recyclers, aggregators, government, and private sector
- Increased readiness for future EPR implementation and enforcement
- Scalable and replicable digital infrastructure for circular economy initiatives
- Contribution to SDGs 12 (Responsible Consumption and Production), 8 (Decent Work), and 13 (Climate Action)

Prototype & MVP Plan

- Demonstration of digital registration and verification of plastic recovery events
- Anchoring of verified recovery records on blockchain with SDG-linked metadata
- Functional dashboards for recovery tracking, certification status, and reporting Initial Plastic Credit issuance linked to certified recovery data
- Collection of structured feedback from recyclers, aggregators, and institutional partners

Expected Outputs

- Blockchain-anchored recovery and certification records
- On-chain certification status with audit-ready metadata
- Operational dashboards for UNDP and partners
- Documented testing and feedback report
- Pilot-ready platform configured for local context

Pilot & Scalability

- Pilot scope: Onboard 150–200 grassroots recyclers, Accredit 2–3 waste management actors (aggregators / facilities), Certify 300–400 tonnes of recovered plastic
- Short term (0–6 months): Localize platform and workflows, Conduct onboarding, training, and verification testing, Publish initial impact dashboards
- Medium term (6–18 months): Launch Plastic Credit issuance and corporate marketplace, Strengthen partnerships and EPR-aligned use cases
- Long term (18+ months): Expand to additional regions and waste streams, Integrate with national EPR systems, Replicate model in other countries

Business & Sustainability Model

- Revenue from Plastic Credit verification and issuance
- Verification and certification fees for accredited waste actors
- Co-financing from sponsor companies and EPR-aligned producers

- Blended finance from climate funds and donor programmes
- Platform access for UNDP provided free of charge during pilot
- Long-term sustainability through market demand for verified impact data

9.10 UNDP Nigeria – Cubid Protocol

Challenge

Rural communities across Nigeria remain underserved by the national electricity grid and continue to rely on expensive, unreliable, and highly polluting energy sources such as diesel generators and kerosene. Although solar mini-grids are a proven technical solution, many fail to scale or remain financially viable due to weak tariff collection systems, fragmented operational data, and limited transparency. Mini-grid operators struggle to demonstrate predictable revenues, governments and development partners lack reliable oversight data, and investors face high verification costs. The core challenge is therefore not electricity generation itself, but the absence of trusted, transparent, and inclusive operating and reporting infrastructure that can support sustainable rural solar mini-grids and unlock access to climate finance.

Solution Overview

Solar Village is a blockchain-enabled operations and reporting platform for rural solar mini-grids. It records electricity production, consumption, and payment data at regular intervals within a local ledger for each mini-grid, and anchors cryptographic proofs of this data to the Cardano blockchain. This creates tamper-evident, auditable records that improve transparency for operators, regulators, and UNDP, while laying the foundation for standards-aligned carbon credit verification and issuance based on verified kilowatt-hour (kWh) data.

Key Blockchain Features

- Per-mini-grid ledgers using Cardano Hydra to record high-frequency energy and payment data
- Daily anchoring of cryptographic proofs to Cardano mainnet for immutability and auditability
- Tamper-evident logging of electricity production and consumption data
- On-chain traceability of payments and settlement flows where applicable
- Tokenization readiness for future energy credits and carbon credit issuance
- Blockchain used as a back-end trust layer, abstracted away from end users

Expected Impact

- Improved transparency and accountability in rural energy operations
- Verifiable reporting of electricity generation, consumption, and payments
- Reduced verification costs for future carbon credit issuance
- Strengthened financial sustainability of solar mini-grid operators
- Increased access to reliable and affordable electricity for rural households
- Contribution to SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure), and SDG 13 (Climate Action)

Prototype & MVP Plan

- Demonstration of automated energy data capture at 15-minute intervals
- Recording of metered data in Hydra heads and anchoring daily proofs on Cardano
- Functional dashboards for households, operators, investors, and oversight actors
- Prepaid electricity top-up flows via web and USSD interfaces
- Initial carbon-readiness logic based on verified energy data
- Structured testing sessions with operators, households, and institutional stakeholders

Expected Outputs

- Tamper-evident on-chain proofs of energy and payment data
- Operational dashboards for operators and UNDP oversight
- Pilot-ready blockchain architecture and documentation
- User testing and feedback documentation
- Carbon-credit-ready data pipelines aligned with emerging standards

Pilot & Scalability

- Pilot scope: Deployment across 23 solar mini-grids identified with REA and AMP, Coverage of 200+ rural households in initial phase
- Short term (0–6 months): Complete system integrations and backend implementation, Deploy Hydra heads and launch pilot dashboards, Execute live pilot with partner mini-grid operators
- Medium term (6–18 months): Expand to additional mini-grids within existing national programmes, Formalize alignment with national and international carbon MRV standards, Introduce structured pricing for operators
- Long term (18+ months): National or multi-region rollout in collaboration with government agencies, Enable carbon credit issuance workflows where approved, Offer Solar Village as a scalable SaaS or digital public infrastructure service

Business & Sustainability Model

- Subscription fees for mini-grid operators post-pilot
- Usage-based or transaction fees tied to metered energy data and reporting services
- Future revenues linked to carbon credit MRV and issuance support
- Institutional service agreements with governments and development partners
- Blended financing approach combining grants, concessional capital, and market revenues

10 Lessons Learned

- **Hackathon as a Strong Starting Point**: The hackathon provided a strong foundation for teams before they deep-dived into the Accelerator and selection phase. It allowed participants to get to know one another and understand each other's capabilities early on, which proved valuable prior to selection for Cohort 2.
- **Centralized Communication Adds Credibility**: Using UNDP AltFinLab as the primary communication channel for UNDP COs helped formalize discussions, reinforce commitment, and ensure participants approached the accelerator with seriousness and professionalism. EMURGO Labs' leadership of the technical component and management of the SMs also contributed to clearly defined structures and responsibilities.
- **Early Engagement with Relevant Stakeholders and Funding Opportunities**: COs should be more strongly encouraged to map and, if possible, engage relevant local stakeholders for pilot implementation during the Accelerator phase. In parallel, both COs and SMs should identify and map potential donors to support the pilot phase.
- **Focus on Small-Scale Pilots**: All teams should be clearly informed that the roadmap and co-development work during the Accelerator should prioritize small-scale pilots. These pilots should require limited funding (maximum of $35,000) and be designed to demonstrate proof of concept and early success, helping to attract additional funding and opportunities for scale-up. This expectation should be clearly communicated from the outset.
- **Co-Financing Expectations**: Both COs and SMs should be aware of the expectation to provide co-financing for pilot activities. COs can leverage existing projects and contribute in-kind support, such as staff time and human resources, as co-financing, without requiring additional funding during the pilot phase. Similarly, SMs should plan to contribute their own resources as part of the co-financing arrangement.
- **Post-Accelerator Support Should Be Defined Early**: Post-Accelerator support should be clearly defined at the beginning of the Accelerator phase so teams are aware of the full program structure and expectations. This should include clarity around ownership and responsibilities.

- **Ownership and Partnership Modalities Must Be Discussed Early**: Early conversations about ownership, roles, responsibilities, and partnership modalities between COs and SMs proved critical. Addressing these topics upfront helped prevent misunderstandings, strengthened accountability, and provided a clear framework for collaboration.